

Conditional Evaluation of Machine Learning Portfolio Strategies: A Functional Inequality Approach

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June 2026

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Abstract

This paper applies the functional inequality test of Li et al. (2025) to evaluate whether machine learning portfolio strategies deliver conditionally superior Sharpe ratios relative to traditional Fama-French factor benchmarks, as a function of the CBOE VIX index. Using daily U.S. equity data spanning January 1998 through December 2025 (7,043 trading days), I test four ML long-short strategies (LASSO, Ridge, Random Forest, LightGBM) against four traditional benchmarks (Market, FF3, FF5, and 1/N equal-weight). No ML strategy conditionally dominates any traditional benchmark in any VIX regime, while the best ML strategy (Ridge) is itself conditionally dominated by traditional strategies in specific market states.

Constructing a confidence set for the most superior strategy across the full universe of eight strategies, only the 1/N equal-weight portfolio survives, with Market, FF3, FF5, and all ML strategies rejected. This provides a conditional counterpart to the unconditional finding of DeMiguel et al. (2009) that the 1/N portfolio is difficult to beat. The baseline (MDS) conclusion holds across $K = 5-8$, CV tuning, subsamples, and transaction costs; under HAC the test marginally rejects, which we report as a stress test of the MDS assumption

1. Introduction

A central question in quantitative portfolio management is whether machine learning methods can generate superior investment performance. Gu et al. (2020) demonstrate that ML models, particularly tree-based ensembles and neural networks, produce out-of-sample stock return predictions that substantially improve upon traditional linear approaches. Their findings, based on a large cross-section of firm characteristics, have motivated a growing literature applying ML to portfolio construction.

These evaluations typically rely on unconditional performance metrics: out-of-sample R^2 , average Sharpe ratios, or cumulative returns. Such averages can mask important heterogeneity across market regimes. A strategy that outperforms in low-volatility expansions but underperforms during crises may appear adequate on average while being unsuitable for an investor who needs reliable performance across all market states. The question of whether ML strategies are “all-weather” portfolios, with conditionally superior performance across the full range of market conditions, requires a formal conditional evaluation framework.

This paper provides such an evaluation by applying the functional inequality test developed by Li et al. (2025). Unlike standard tests of predictive ability such as Hansen (2005), which evaluate performance unconditionally, the functional inequality test evaluates whether a benchmark strategy is dominated conditional on observable state variables. The null hypothesis states that the benchmark is not conditionally dominated by any alternative strategy across the full support of the conditioning variable. Rejection implies that there exist specific market conditions under which at least one ML strategy delivers a statistically significantly higher conditional Sharpe ratio than the benchmark.

The test of Li et al. (2025) generalizes earlier work on conditional superior predictive ability (Giacomini & White, 2006; Li et al., 2022) to accommodate nonlinear functionals of conditional moments, including the Sharpe ratio, which involves the ratio of the conditional mean to the conditional standard deviation. The test is uniformly valid under general forms of time series dependence, which is essential for financial applications. The implementation uses orthonormal Legendre polynomial basis functions for nonparametric estimation, shape-constrained conditional variance estimation via quadratic programming, and a two-step Bonferroni-type critical value procedure following Romano et al. (2014).

The conditioning variable is the CBOE VIX index. This choice follows the empirical application in Li et al. (2025), who recommend the VIX as a default conditioning variable for portfolio evaluation. The analysis uses daily returns on 25 size and book-to-market sorted portfolios from Kenneth French’s data library, covering 7,043 trading days from January 1998 through December 2025. I construct four traditional benchmarks and four ML long-short strategies trained monthly on a rolling 60-month window.

Two main findings emerge. First, no ML strategy conditionally dominates any traditional benchmark under any VIX regime. The functional inequality test statistics range from 2.34 to 2.67, com-

fortably below the critical values of approximately 3.0. In the reverse direction, the best-performing ML strategy (Ridge, annualized Sharpe ratio 0.06) is rejected as conditionally superior over traditional strategies (test statistic 3.67, critical value 3.00). Second, constructing a confidence set for the most superior strategy by rotating the benchmark role across all eight strategies reveals that the 1/N equal-weight portfolio is the only strategy whose conditional superiority cannot be rejected. This extends the unconditional evidence of DeMiguel et al. (2009) by establishing a conditional version: the 1/N portfolio is the only “all-weather” strategy in the Sharpe ratio sense across all VIX states.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and portfolio construction. Section 4 presents the testing methodology. Section 5 reports the empirical results. Section 6 conducts robustness checks. Section 7 concludes.

2. Related Literature

This paper sits at the intersection of three strands of research: machine learning in asset pricing, the naive-diversification debate, and the econometric theory of conditional forecast evaluation. I outline each strand below and then state where the present paper fits.

2.1 Machine Learning in Asset Pricing

The ML-in-finance literature, following Gu et al. (2020), has focused primarily on prediction accuracy, measuring how well ML models forecast individual stock returns out of sample using large panels of firm characteristics. The strong predictive performance documented in this literature, particularly for tree-based ensembles and neural networks, has motivated practitioners to adopt ML-based portfolio construction. However, the evaluation of these portfolios has remained overwhelmingly unconditional. Strategies are compared on their average Sharpe ratios, cumulative wealth, or mean absolute errors, without asking whether the performance advantage holds uniformly across market conditions. The question of conditional performance, whether ML strategies maintain their edge when volatility spikes or the yield curve inverts, has received almost no formal treatment.

2.2 Naive Diversification and the 1/N Puzzle

A parallel strand of the portfolio literature, originating with DeMiguel et al. (2009), questions whether sophisticated quantitative methods improve upon simple heuristics. Their comparison of fourteen portfolio strategies across seven datasets found that the naive 1/N equal-weight rule is difficult to beat after accounting for estimation error in the inputs to mean-variance optimization. This finding has set a high bar for any quantitative method, whether based on classical optimization or on ML. The result is unconditional in nature: DeMiguel et al. (2009) compare average performance metrics across the full sample. Whether the 1/N portfolio’s advantage is uniform across market states, or whether it too is conditionally dominated in specific regimes, remains unresolved prior to the present study.

2.3 Conditional Forecast Evaluation

The forecast evaluation literature provides the formal tools for comparing competing strategies. Unconditional tests, such as the Superior Predictive Ability test of Hansen (2005), compare average loss functions and account for the multiplicity problem that arises when screening many candidate models. Giacomini and White (2006) introduce the conditional dimension by allowing forecast comparisons to depend on observable state variables, capturing the possibility that one model may outperform in some states but not others. Li et al. (2022) extend this to the multi-alternative setting with a sup-type test statistic and a single conditioning variable.

The present paper uses the most recent generalization: Li et al. (2025) develop a functional inequality test that accommodates nonlinear functionals of conditional moments, including the Sharpe ratio, with uniform validity under general mixingale-type dependence. Their framework handles the econometric complications specific to the Sharpe ratio setting, namely shape-constrained conditional variance estimation and the generated dependent variable that arises from normalizing returns by an estimated conditional standard deviation. The empirical application in Li et al. (2025) evaluates four classical portfolio strategies (1/N, three variants of global minimum variance portfolios) across twelve datasets from the U.S. equity market.

2.4 Positioning of This Paper

This paper contributes at the intersection of these three strands. It takes the empirical question from the ML-in-finance literature (do ML strategies outperform?), the benchmark from the naive-diversification literature (can anything beat 1/N?), and the testing framework from the conditional evaluation literature (does outperformance hold across all market states?). The specific contribution beyond Li et al. (2025) is empirical: their paper evaluates classical portfolio construction methods against each other, whereas the present paper introduces ML-based strategies into the comparison. The finding that the 1/N portfolio is the only strategy surviving the conditional superiority test, including over ML alternatives, connects the DeMiguel et al. (2009) result to the conditional evaluation framework in a way that has not been documented previously.

3. Data and Portfolio Construction

3.1 Data Sources and Sample

Daily returns on 25 portfolios sorted on size and book-to-market are obtained from Kenneth French’s data library. These portfolios are formed at the end of each June by intersecting five NYSE size quintiles with five NYSE book-to-market quintiles. The Fama and French (2015) five factors (Market, SMB, HML, RMW, CMA) and the momentum factor are obtained at both daily and monthly frequencies from the same source. The CBOE VIX index is obtained at daily frequency from the Federal Reserve

Economic Data (FRED) database.

The sample runs from January 1998 through December 2025, yielding $n = 7,043$ trading days. The start date is determined by VIX availability at daily frequency. The sample covers the dot-com bust (2000–2002), the Global Financial Crisis (2007–2009), the low-volatility period of 2013–2017, the COVID-19 crash and recovery (2020), and the post-pandemic tightening cycle (2022–2025), providing variation across distinct volatility regimes.

3.2 Conditioning Variable: The VIX

The VIX measures the implied volatility of S&P 500 options over the next 30 days and serves as the primary conditioning variable throughout the analysis. Over the sample period, the VIX ranges from approximately 9 to 83, with a median of 17.5 and a mean of 20.4. Its distribution is right-skewed, reflecting the well-known observation that volatility spikes are more common than volatility collapses. The transformation applied in Section 4.2 (log followed by normal CDF mapping) symmetrizes this distribution and maps it onto $[-1, 1]$, which is desirable for the Legendre polynomial sieve.

Roughly 65% of trading days fall in a low-VIX regime (VIX below 20), approximately 25% in a moderate regime (VIX between 20 and 30), and about 10% in a high-VIX regime (VIX above 30). The conditional Sharpe ratio test draws its statistical power from the variation across these regimes: if ML strategies outperform in one regime but not others, the functional inequality test can detect the discrepancy even when unconditional metrics show no difference.

3.3 Traditional Benchmark Strategies

Four traditional strategies are constructed from the factor and portfolio data. The *Market* strategy is the daily market excess return (Mkt-RF). *FF3* and *FF5* are equal-weighted combinations of the returns to the three and five Fama-French factor-mimicking portfolios, respectively. Since each factor return corresponds to a tradable portfolio (zero-cost for the non-market factors), these composites are themselves implementable strategies, representing a naive equal-weight tilt across the canonical factor set, a factor-level counterpart to the 1/N rule applied across the sorted portfolios. The *EqWeight* strategy is the 1/N equal-weighted average of the 25 sorted portfolio returns.

3.4 ML-Based Portfolio Strategies

The ML strategies are constructed at monthly frequency via a rolling-window procedure. The pipeline at each month-end rebalance t proceeds in four steps. First, a panel dataset is assembled from the prior 60 months: for each of the 25 portfolios and each month in the window, one observation consists of the 11 features described below and the target variable (next-month portfolio return), yielding $25 \times 60 = 1,500$ panel observations. Second, all features are standardized to zero mean and unit variance within the training window. Third, the model is fit and predictions are generated for all 25 portfolios

at month $t+1$. Fourth, the five portfolios with the highest predicted returns receive equal long weights of $1/5$ and the five with the lowest receive equal short weights of $-1/5$; these weights are held constant through month $t+1$, and daily returns are computed by applying them to the daily returns of the underlying portfolios.

The feature set consists of three lagged own-return variables (the portfolio’s return at lags 1, 2, and 3), seven lagged factor values (the five Fama-French factors, momentum, and the risk-free rate, all at lag 1), and the cross-sectional percentile rank of the most recent month’s return. The rank feature captures relative momentum: portfolios that performed well relative to their peers receive a high rank value. The 60-month training window balances estimation precision against responsiveness to structural change; shorter windows (e.g., 36 months) would yield noisier estimates, while longer windows (e.g., 120 months) would dilute the relevance of recent data.

Four models are considered. LASSO (ℓ_1 -penalized regression, $\alpha = 0.001$) performs feature selection by shrinking coefficients toward zero. Ridge (ℓ_2 -penalized regression, $\alpha = 1.0$) retains all features but penalizes large coefficients. Random Forest (100 trees, maximum depth 5, minimum leaf size 10) captures nonlinear feature interactions through ensemble averaging. LightGBM (100 boosting rounds, maximum depth 4, learning rate 0.05) sequentially fits trees to the residuals of prior iterations. The regularization parameters for LASSO and Ridge are set ex ante; a robustness check in Section 6.4 replaces them with cross-validated selection.

Table 1 reports annualized performance. All four ML strategies produce near-zero or negative Sharpe ratios, with Ridge the strongest at 0.06. Figure 1 shows the cumulative wealth paths. The traditional strategies accumulate wealth monotonically over the sample, while the ML strategies oscillate around their starting values.

Table 1: Annualized Performance, January 1998 to December 2025

Strategy	Ann. Mean	Ann. Vol	Sharpe	N
Market	0.088	0.196	0.452	7,043
FF3	0.036	0.089	0.408	7,043
FF5	0.033	0.055	0.596	7,043
EqWeight	0.117	0.215	0.547	7,043
ML LASSO	-0.001	0.120	-0.008	7,043
ML Ridge	0.007	0.115	0.061	7,043
ML Random Forest	0.001	0.147	0.005	7,043
ML LightGBM	-0.017	0.122	-0.135	7,043

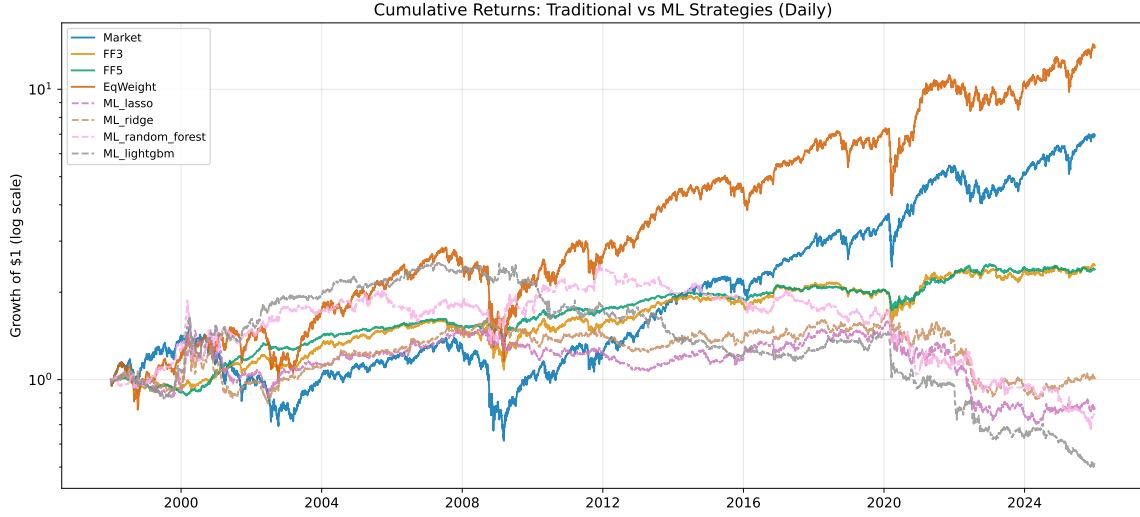


Figure 1: Cumulative wealth paths (log scale) for traditional and ML strategies, 1998–2025. Traditional benchmarks are shown in solid lines; ML strategies in dashed lines.

4. Methodology: The Functional Inequality Test

This section describes the functional inequality test of Li et al. (2025) as applied to the conditional Sharpe ratio setting. I follow the notation of the original paper and expand on the key econometric issues that arise in practice.

4.1 The Testing Problem

Let $r_{j,t+1}$ denote the excess return of strategy j over period $[t, t+1]$, and let X_t denote a conditioning variable observed at time t . The conditional mean and variance of $r_{j,t+1}$ given $X_t = x$ are $\mu_j(x) = \mathbb{E}[r_{j,t+1} \mid X_t = x]$ and $\sigma_j^2(x) = \text{Var}(r_{j,t+1} \mid X_t = x)$. The conditional Sharpe ratio of strategy j at state x is

$$\text{SR}_j(x) = \frac{\mu_j(x)}{\sigma_j(x)}. \quad (1)$$

Index the benchmark by $j = 0$ and the alternatives by $j \in \mathcal{J} = \{1, \dots, J\}$. The null hypothesis of conditional benchmark superiority is

$$H_0 : \text{SR}_0(x) \geq \text{SR}_j(x) \quad \text{for all } j \in \mathcal{J}, x \in \mathcal{X}, \quad (2)$$

where $\mathcal{X}$ is the support of the conditioning variable. Rejection implies the existence of at least one alternative and one conditioning state at which the alternative's Sharpe ratio exceeds the benchmark's. This is a functional inequality: the constraint must hold pointwise for every value of x , not merely on average.

4.2 Sieve Construction and the Conditioning Variable Transform

The unknown conditional Sharpe ratio functions are estimated nonparametrically using a series (sieve) approach. The approximating basis consists of orthonormal Legendre polynomials $\psi_0(x), \psi_1(x), \dots, \psi_{K-1}(x)$ on $[-1, 1]$. Orthonormality means $\int_{-1}^1 \psi_i(x) \psi_j(x) dx/2 = \mathbf{1}\{i = j\}$, which ensures the Gram matrix $\hat{Q} = n^{-1} \sum_{t=1}^n \psi(X_t^*) \psi(X_t^*)^\top$ is well-conditioned when the transformed conditioning variable X_t^* is approximately uniformly distributed on $[-1, 1]$.

To achieve this approximate uniformity, the raw VIX is transformed in two steps. First, because the VIX distribution is right-skewed, I take its logarithm: $\tilde{X}_t = \log(X_t)$. Second, the log-VIX is standardized and mapped through the standard normal CDF: $X_t^* = 2 \Phi((\tilde{X}_t - \bar{\mu})/\bar{\sigma}) - 1$, where $\bar{\mu}$ and $\bar{\sigma}$ are the sample mean and standard deviation of $\tilde{X}_t$. This maps the observed conditioning variable onto $[-1, 1]$ with an approximately uniform marginal, under the assumption that $\tilde{X}_t$ is approximately Gaussian, which holds reasonably well for log-VIX over the sample period.

The sieve dimension is $K = \max\{4, \lfloor 1.2 n^{1/5} \rfloor\}$, following Li et al. (2022). For $n = 7,043$, this gives $K = 7$, allowing the sieve to fit up to a sixth-order polynomial in the transformed conditioning variable. Section 6.2 examines sensitivity to this choice.

4.3 Nonparametric Estimation and the Generated-Variable Problem

The conditional mean function is estimated by sieve regression:

$$\hat{\mu}_j(\cdot) = \psi(\cdot)^\top \hat{Q}^{-1} \left(n^{-1} \sum_{t=1}^n \psi(X_t^*) r_{j,t+1} \right), \quad \hat{Q} = n^{-1} \sum_{t=1}^n \psi(X_t^*) \psi(X_t^*)^\top. \quad (3)$$

The conditional variance $\sigma_j^2(\cdot)$ is estimated by a sieve regression on the squared residuals $(r_{j,t+1} - \hat{\mu}_j(X_t^*))^2$, subject to a positivity constraint. Because the unconstrained sieve estimate can produce negative fitted values at some observed points (a finite-sample artifact of polynomial approximation), a quadratic programming step enforces $\hat{\sigma}_j^2(X_t^*) \geq \sigma_{j,n}^2$ for all t , where $\sigma_{j,n}^2 = \min_t \tilde{\sigma}_j^2(X_t^*) / \log(n)$ is a data-driven lower bound based on a preliminary consistent estimator $\tilde{\sigma}_j^2$ (Equation 3.25 of Li et al. (2025)). The $\log(n)$ denominator ensures the constraint becomes asymptotically unbinding, so the shape-restricted estimator inherits the asymptotic properties of its unrestricted counterpart.

The conditional Sharpe ratio is then estimated by normalizing returns by the estimated conditional standard deviation: $\hat{r}_{j,t+1}^* = r_{j,t+1} / \hat{\sigma}_j(X_t^*)$, followed by sieve regression of $\hat{r}_{j,t+1}^*$ on $\psi(X_t^*)$. This normalization creates a generated dependent variable: the “data” entering the Sharpe ratio regression depends on the first-step estimator $\hat{\sigma}_j$, which is itself estimated from the same sample. Ignoring this dependence would understate the sampling variability of the Sharpe ratio estimator, because the estimation error in $\hat{\sigma}_j$ propagates into $\hat{r}_{j,t+1}^*$ and hence into the sieve coefficients of the Sharpe ratio regression.

To account for this, the covariance matrix $\hat{\Omega}$ is constructed from the influence function

$$\hat{v}_{j,t+1} = \psi(X_t^*) \hat{\epsilon}_{j,t+1} - \hat{\Upsilon}_j \psi(X_t^*) \hat{\eta}_{j,t+1}, \quad (4)$$

where $\hat{\epsilon}_{j,t+1} = \hat{r}_{j,t+1}^* - \widehat{\text{SR}}_j(X_t^*)$ is the Sharpe ratio regression residual, $\hat{\eta}_{j,t+1} = (r_{j,t+1} - \hat{\mu}_j(X_t^*))^2 - \hat{\sigma}_j^2(X_t^*)$ is the variance regression residual, and $\hat{\Upsilon}_j$ is a $K \times K$ matrix that captures the sensitivity of the Sharpe ratio sieve coefficients to changes in the variance estimate. The first term of $\hat{v}_{j,t+1}$ is the standard regression influence; the second term is the generated-variable correction. Omitting the second term would produce standard errors that are too small, biasing the test toward over-rejection.

4.4 Dependence Structure and the MDS Assumption

The influence function $\hat{v}_{j,t+1}$ is used to estimate the long-run covariance matrix Ω_v . If $\{v_{j,t+1}\}$ forms a martingale difference sequence (MDS), the sample covariance $n^{-1} \sum_t \hat{v}_{t+1} \hat{v}_{t+1}^\top$ is a consistent estimator of Ω_v . Under a no-arbitrage argument, daily excess returns are plausibly MDS: predictable returns would be arbitrated away. However, the influence function $\hat{v}_{j,t+1}$ involves nonlinear transformations of returns (the squared residuals $\hat{\eta}_{j,t+1}$, the normalized returns $\hat{r}_{j,t+1}^*$), and these transformations do not necessarily inherit the MDS property even if the returns themselves are MDS. For this reason, I report the baseline results using the sample covariance (the MDS specification) and provide Newey-West HAC estimates as a robustness check in Section 6.3. The theoretical framework of Li et al. (2025) accommodates general mixingale-type dependence, which subsumes both cases.

4.5 Test Statistic and the Two-Step Critical Value

The test statistic is a one-sided sup- t statistic:

$$\hat{T}_n = \max \left(\max_{j \in \mathcal{J}} \sup_{x \in \mathcal{X}} \frac{\sqrt{n} \hat{h}_j(x)}{\hat{s}_j(x)}, 0 \right), \quad (5)$$

where $\hat{h}_j(x) = \widehat{\text{SR}}_j(x) - \widehat{\text{SR}}_0(x)$ and $\hat{s}_j(x)$ is the pointwise standard error from the diagonal block of $\hat{\Omega}$ corresponding to alternative j . The supremum is evaluated over 950 grid points on $\mathcal{X}$, trimmed at the 2.5% and 97.5% quantiles of X_t^* to avoid noisy boundary estimates. The truncation at zero reflects the one-sided nature of the null: only positive differentials (alternative outperforms benchmark) contribute to the test statistic.

A naive critical value would be the $(1 - \alpha)$ quantile of the simulated Gaussian process under the least favorable configuration (setting all functional inequalities to equality). This approach is valid but conservative: when some inequalities are strictly slack (the benchmark strongly dominates certain alternatives), the least favorable critical value is too large, reducing the power of the test.

The two-step procedure of Romano et al. (2014), adapted to the functional inequality setting by Li et al. (2025), improves power by estimating which inequalities are slack and adjusting the critical value

accordingly. In step one, a uniform upper confidence bound is constructed at level $1 - \rho$:

$$\hat{u}_j(x) = \min\{\hat{h}_j(x) + n^{-1/2}\hat{s}_j(x)\hat{q}_{1-\rho}, 0\},$$

where $\hat{q}_{1-\rho}$ is the $(1 - \rho)$ quantile of the simulated Gaussian maximum. The truncation at zero ensures the bound respects the null hypothesis. In step two, the critical value is the $(1 - \alpha + \rho)$ quantile of the adjusted simulated statistic incorporating $\hat{u}_j$. The overall rejection probability under the null is bounded by $\rho + (\alpha - \rho) = \alpha$: the first step contributes at most ρ to the false rejection probability, and the second step contributes at most $\alpha - \rho$. I use $\rho \in \{0.001, 0.005\}$ following the recommendation of Li et al. (2025) and $\alpha = 0.05$, with 10,000 Gaussian simulation draws.

A confidence set for the most superior (CSMS) strategy is constructed by rotating the benchmark role across all strategies. A strategy enters the CSMS if the null of its conditional superiority over all others is not rejected at the α level.

5. Empirical Results

5.1 Bilateral Tests: Traditional versus ML

Table 2 reports the functional inequality test for four benchmark configurations. In Tests 1–3, each traditional benchmark is evaluated against the four ML alternatives. In Test 4, the best ML strategy by unconditional Sharpe ratio (Ridge, at 0.06) is evaluated against the four traditional strategies.

Table 2: Functional Inequality Test: Bilateral Comparisons (5% level, $\rho = 0.001$)

Test	Benchmark	Alternatives	$\hat{T}_n$	CV	Reject
1	Market	ML (all 4)	2.337	3.010	No
2	FF5	ML (all 4)	2.521	3.048	No
3	EqWeight	ML (all 4)	2.675	3.036	No
4	ML Ridge	Traditional (all 4)	3.668	2.998	Yes

The pattern is asymmetric. Tests 1–3 fail to reject: no ML strategy’s conditional Sharpe ratio exceeds any traditional benchmark’s at any VIX level. Test 4 rejects: there exist VIX states in which at least one traditional strategy conditionally outperforms the best ML strategy. The directional conclusion is that traditional strategies conditionally dominate ML, and not the reverse.

Figure 2 displays the estimated conditional Sharpe ratio functions from Test 1. The Market’s conditional Sharpe ratio (solid line) is high in low-VIX environments (approximately 0.5 when VIX is near 10) and declines sharply as VIX increases, turning negative above approximately VIX 25. This pattern reflects several well-documented mechanisms: the leverage effect (negative correlation between

returns and volatility innovations), time-varying risk premia that compress during stress, and flight-to-safety flows that depress equity returns during high-volatility episodes. The ML strategies' curves (dashed lines) are flatter and remain near zero across most of the conditioning space. Their conditional Sharpe ratios do not respond to VIX in a systematic way, consistent with the interpretation that the ML models' predictions contain little signal about state-dependent expected returns or risk.

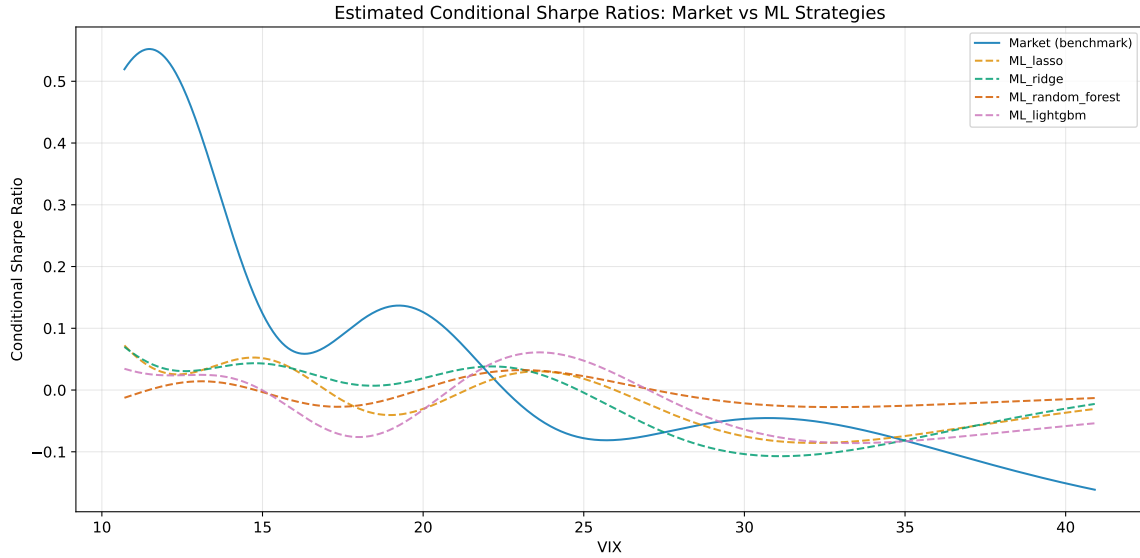


Figure 2: Estimated conditional Sharpe ratios as a function of VIX for the Market benchmark and four ML strategies. The evaluation grid is trimmed at the 2.5% and 97.5% quantiles of the transformed VIX.

5.2 Confidence Set for the Most Superior Strategy

Table 3 reports the CSMS analysis. Each strategy is used as the benchmark against all seven remaining strategies. A strategy enters the CSMS if the null of its conditional superiority is not rejected.

Table 3: Confidence Set for the Most Superior (CSMS), 5% level, $\rho = 0.001$

Strategy	$\hat{T}_n$	CV	In CSMS
Market	3.398	3.181	No
FF3	4.041	3.242	No
FF5	3.515	3.244	No
EqWeight	3.198	3.212	Yes
ML LASSO	3.376	3.232	No
ML Ridge	3.668	3.232	No
ML Random Forest	4.971	3.193	No
ML LightGBM	4.798	3.207	No

The 1/N equal-weight portfolio is the only strategy in the confidence set. All four ML strategies are excluded, confirming the bilateral finding. But the CSMS analysis reveals a finer result: Market, FF3, and FF5 are also excluded. Their conditional superiority is rejected not because of ML alternatives, but because other traditional strategies (plausibly EqWeight) conditionally dominate them in certain VIX regimes. The CSR plot in Figure 2 is consistent with this: the Market’s conditional Sharpe ratio drops sharply below zero for VIX above 25, while the EqWeight portfolio, as a diversified average across 25 size/BM cells, is more stable.

The contrast between FF5’s unconditional and conditional standing is worth noting. FF5 has the highest unconditional Sharpe ratio in the sample (0.596), yet it is excluded from the CSMS. Its conditional Sharpe ratio function, while high on average, falls below that of EqWeight in specific VIX regimes, producing a test statistic (3.515) that exceeds the critical value (3.244). This is precisely the type of distinction that unconditional evaluation cannot detect: a strategy can have the highest average performance while still being conditionally dominated in particular market states. The 1/N portfolio, with a lower unconditional Sharpe of 0.547, is more uniform across the conditioning space and survives the conditional test.

The EqWeight result is borderline ($\hat{T}_n = 3.198$ versus $CV = 3.212$, a margin of 0.014) and should be interpreted with appropriate caution. Nonetheless, the qualitative finding that the 1/N portfolio is the most conditionally adequate strategy in this universe provides a conditional counterpart to the unconditional evidence of DeMiguel et al. (2009).

6. Robustness Checks

6.1 Pairwise Tests

Table 4 reports pairwise tests of Market versus each ML strategy individually. All four fail to reject, with test statistics between 1.71 and 2.34. The pairwise statistics are lower than the joint test statistic (2.337 in Test 1) because the joint test takes the maximum across all four alternatives, while each pairwise test considers a single alternative in isolation.

Table 4: Pairwise Tests: Market vs. Each ML Strategy ($\rho = 0.001$)

Alternative	$\hat{T}_n$	CV	Reject
ML LASSO	1.709	2.695	No
ML Ridge	1.807	2.696	No
ML Random Forest	2.337	2.677	No
ML LightGBM	1.742	2.676	No

6.2 Sensitivity to Sieve Dimension

Table 5 varies the sieve dimension from $K = 4$ to $K = 8$ for Test 1. Non-rejection holds for $K = 5$ through $K = 8$. At $K = 4$ the test marginally rejects ($\hat{T}_n = 2.99$, $CV = 2.79$). This is consistent with a general pattern: smaller K produces smoother sieve estimates that can detect broad departures from the null, while larger K allows the sieve to fit local variation that may absorb noise rather than signal. The recommended $K = 7$ lies within the non-rejection range, and the findings are not sensitive to this choice for $K \geq 5$.

Table 5: Sensitivity to K : Market vs. ML ($\rho = 0.001$)

K	$\hat{T}_n$	CV	Reject
4	2.991	2.787	Yes
5	2.687	2.855	No
6	2.474	2.922	No
7	2.337	3.010	No
8	2.141	3.044	No

6.3 HAC Covariance

The baseline analysis assumes the influence function $\{v_{t+1}\}$ forms an MDS, in which case the sample covariance is an appropriate estimator for Ω_v . The MDS assumption for daily excess returns is mo-

tivated by no-arbitrage: predictable expected returns would be traded away in liquid equity markets. However, as discussed in Section 4.4, the influence function involves nonlinear transformations of returns (squared residuals, normalized returns), and these transformations can induce serial dependence even when the raw returns are MDS. This is not a failure of the test framework, which accommodates general dependence through the mixingale theory of Li et al. (2025), but it does mean that the choice of covariance estimator affects the finite-sample test outcome.

Replacing the sample covariance with the Newey-West estimator (with automatic lag selection $\ell = \lfloor 1.2 \cdot n^{1/3} \rfloor$), the test statistic for Test 1 increases from 2.34 to 3.16, producing a marginal rejection ($CV = 3.01$). The rise in the test statistic implies that, at the binding region of the conditioning space, the HAC-estimated long-run variance of the influence function is smaller than the MDS sample variance, which lowers $\hat{\sigma}^2(\mathbf{x})$ and inflates the studentized statistic. Positive autocovariance would instead raise the long-run variance and the standard errors, lowering the statistic, and so cannot account for the observed increase. Whether this marginal rejection represents a genuine signal or a finite-sample artifact of the HAC estimator’s own variance is difficult to adjudicate from the data alone. The baseline MDS specification is maintained as the primary result, following Li et al. (2025), and the HAC result is reported as a transparent stress test of this assumption.

6.4 Cross-Validated Hyperparameter Tuning

A natural concern is that the weak ML performance may reflect suboptimal regularization. The baseline LASSO and Ridge use fixed regularization parameters ($\alpha = 0.001$ and $\alpha = 1.0$) chosen ex ante. I re-estimate both using 5-fold cross-validation within each rolling training window: LassoCV selects α by minimizing cross-validated mean squared error, and RidgeCV searches a logarithmic grid from 10^{-3} to 10^3 . The tree-based models are unchanged.

Table 6: Fixed vs. Cross-Validated Tuning: Annualized Sharpe Ratios

Model	Fixed	CV-Tuned	Difference
LASSO	−0.008	−0.325	−0.317
Ridge	0.061	0.067	+0.006
RF	0.005	0.005	0.000
LightGBM	−0.135	−0.135	0.000

CV tuning does not improve ML performance (Table 6). Ridge is essentially unchanged. LASSO deteriorates substantially, as the CV procedure selects more aggressive shrinkage in some windows, setting more coefficients to zero and reducing the strategy’s ability to differentiate across portfolios. The CSPA test on the CV-tuned strategies continues to fail to reject ($\hat{T}_n = 2.97$, $CV = 3.05$). The weak conditional performance of the ML strategies is not an artifact of suboptimal hyperparameter

selection.

6.5 Subsample Stability

Splitting the sample at January 2010, the non-rejection of Market conditional superiority over ML holds in both subsamples: $\hat{T}_n = 2.78$ (CV = 2.92) for the pre-2010 period ($n = 3,019$, $K = 5$) and $\hat{T}_n = 1.20$ (CV = 2.90) for the post-2010 period ($n = 4,024$, $K = 6$). The substantially weaker test statistic in the post-2010 subsample reflects the even poorer relative performance of ML strategies in recent years, when the low-volatility environment and momentum reversals made lagged-return features particularly unreliable.

6.6 Transaction Costs

Applying a proportional transaction cost of 10 basis points per unit of monthly turnover to the ML strategies (traditional benchmarks are assumed to have negligible turnover, as the Market is passive and the factor and equal-weight portfolios require minimal rebalancing), the test result for Market versus ML is unchanged: $\hat{T}_n = 2.23$, CV = 3.00, no rejection.

7. Conclusion

This paper tests whether ML portfolio strategies deliver conditionally superior Sharpe ratios relative to traditional Fama-French benchmarks across all VIX states. Two findings stand out.

The bilateral tests establish that traditional strategies conditionally dominate ML strategies, and not the reverse. No ML strategy's conditional Sharpe ratio exceeds any traditional benchmark's at any VIX level, while the best ML strategy is itself conditionally dominated by traditional strategies in specific regimes. These results are stable across pairwise and joint tests, across sieve dimensions $K = 5$ through $K = 8$, under cross-validated hyperparameter tuning, across pre- and post-2010 subsamples, and after adjusting for transaction costs. Under the alternative HAC covariance specification the test marginally rejects, which we report as a stress test of the MDS assumption rather than as the primary result.

The CSMS analysis adds a second layer: even among traditional strategies, only the 1/N equal-weight portfolio qualifies as conditionally superior across the full strategy universe. The rejections of Market, FF3, and FF5 from the CSMS are driven by conditional dominance within the set of traditional strategies in specific VIX regimes, not by ML alternatives. This finding provides a conditional counterpart to DeMiguel et al. (2009), who documented unconditionally that the 1/N portfolio is difficult to beat.

Several limitations apply. The ML strategies use a simple feature set applied to 25 sorted portfolios, not the hundreds of firm-level characteristics and thousands of individual stocks used in Gu et al. (2020). More complex implementations could yield different conclusions. The analysis conditions on

a single variable (VIX); other state variables could reveal additional patterns. The EqWeight CSMS membership is borderline ($\hat{T}_n = 3.198$, $CV = 3.212$) and should be interpreted as suggestive rather than definitive.

Future extensions could apply the functional inequality framework to ML strategies built from individual stock-level data, use the modified Sharpe ratio with Value-at-Risk as the risk measure (accommodated within Li et al. (2025)), or extend the conditioning analysis to multiple state variables.

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